

EDUCATION

Degree	Institute	CGPA/Percentage	Year
B.S. Electronics and Communication	Indian Institute of Science Education and Research, Bhopal	8.44/10	2026
Senior Secondary	Alard Public School, CBSE Board	82.0%	2022
Secondary	Spicer Higher Secondary School, ICSE Board	92.83%	2020

EXPERIENCE

Position	Institute/Company	Year	Advisor
Research Intern	Indian Institute of Technology, Kanpur	May 2025 - July 2025	Dr. Soumya Ranjan Sahoo
Research Intern	Indian Institute of Science Education and Research, Bhopal	March 2024 - Present	Dr. Arijit Sen

PROJECTS

- **Three Dimensional UAV Formation Control on Time-Varying Inclines**

May 2025 - July 2025

Research Project / Dr.Soumya Ranjan Sahoo and Dr.Arijit Sen

– Formulated a nonlinear Lyapunov-based control law to stabilize UAV formation geometry, enabling precise 3D tracking of moving target on surfaces with time-varying inclination.

– Proved the asymptotic stability of the controller mathematically and validated its performance via extensive MATLAB simulations.
- **Cooperative Localization to Maximize Target Information in Three Dimensions**

March 2024 - Present

Research Project / Dr.Arijit Sen

– Architected a Cooperative Localization framework for 3D target estimation.

– Derived non-linear control laws that drive a multi-UAV agents into optimal geometric configurations by maximizing the Fisher Information Matrix (FIM).

– Currently implementing the closed-loop simulation in MATLAB to benchmark the proposed strategy against static formation baselines.
- **Safe and Autonomous Docking of an Autonomous Underwater Vehicle**

Sept 2025-Nov 2025

Intelligent Robotics ECS 418 Course Project/ Prof.Sujit PB

– Investigated Safety-Critical Control methods to solve the constrained docking problem for underactuated AUVs.

– Formulated a non-linear control framework integrated with a extended kalman filter for precise estimation of docking in realistic conditions.

– Validated the controller in MATLAB simulations, demonstrating that the AUV respects safety constraints even when the nominal controller attempts to violate them, ensuring collision-free docking.

PUBLICATIONS

- **S. Shaha**, A. Sen and S.R. Sahoo, Three Dimensional UAV Formation Control on Time-Varying Incline (under preparation)

• **S. Shaha**, G.V.S. Sriram and A. Sen, Cooperative Localization to Maximize Target Information in Three Dimensions using Agents in Formation (under preparation)

• A.Shukla, **S. Shaha**, and A. Sen, Multi-Agent Formation Control and Target Tracking on the Hyperboloid Manifold (under preparation)

TECHNICAL SKILLS

- **Programming:** Python (Numpy, Matplotlib), LaTeX

• **Software:** MATLAB, Simulink, AutoCAD

KEY COURSES TAKEN

Intelligent Robotics, Modern Control Systems, Optimization Techniques, Robotic Perception, Numerical Analysis, Multi-agent Reinforcement Learning

AWARDS

- **IOQM Awardee**, Qualified Indian Olympiad Qualifier in Mathematics

2021
- **Indian Control Conference 11 Student Support Award**, Held at IISc Bengaluru

2025
- **Indian Control Conference 10 Student Support Award**, Held at IISER Bhopal

2024
- **SURGE Internship Program**, IIT Kanpur

2025